Next Steps and Reflections

Personal Learning and Reflection

- Gained a deeper understanding of quantum computing by transforming abstract concepts into visual representations.
- Discovered how programming, design, and scientific thinking can work together to explain complex ideas.
- Found the most rewarding moment was successfully generating ZX-style diagrams that represented real quantum operations.
- Developed confidence in independent learning, problem-solving, and creative experimentation through the project.

Future Development and Improvements

- Plans to expand the visualiser to support more advanced concepts such as multi-qubit entanglement.
- Intends to introduce interactive features, including draggable spiders, user-created diagrams, and animated phase changes.
- Would like to integrate the visualiser directly with Qiskit, enabling automatic conversion from quantum circuits to ZX-diagrams.
- Aims to develop the project into a more comprehensive educational tool for schools and universities.

Long-Term Potential and Career Impact

- Identified the need for a more polished interface and improved performance to make the software suitable for wider use.
- Suggested creating a web-based version using technologies such as JavaScript or React to improve accessibility.
- Recognised the value of collaborating with educators to ensure the visualisations support effective classroom learning.
- Inspired a continued interest in quantum computing and educational software development, demonstrating how creativity, persistence, and curiosity can make advanced STEM subjects more engaging and accessible.